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	The overall objective of this research project is to develop noninvasive sensing technology for in situ chemical monitoring of critical biochemical processes. Specific aims of this project are to 1) develop dedicated NIR spectroscopic instrumentation capable of noninvasive in situ monitoring of critical clinical analytes, specifically glucose and lactate, 2) develop instrumentation for the in situ monitoring of soluble protein in real-time during crystal formation, and 3) to evaluate the robustness and durability of this technology for function under non-laboratory conditions, such as on board the International Space Station or during long-term space flights. The specific aim of this research project is to develop near infrared spectroscopic methods for the measurement of key system analytes within complex biological matrixes. The principal objective is to develop the instrumentation for measuring glucose and lactate within the human body for noninvasive health monitoring. A secondary objective is the development instrumentation for monitoring and controlling the rate of protein crystal growth as a means to optimize the formation of high quality protein crystals for crystallographic purposes. For both objectives, we are focused on the development of novel spectroscopic hardware that		

Task Description:

provides sufficient signal-to-noise for meaningful analytical measurements in complex systems. An additional aim of this work is to produce instrumentation with sufficient robustness for practical implementation in either a clinical setting or on board space craft during long-term space flights.
Two optical designs are under consideration. The first is based on a set of four band-pass filters and the second is based on an acousto-optic tunable filter (AOTF). The former offers a robust, all solid state instrument design but requires a priori knowledge of the bandpass requirements of each filter. The second approach provides greater flexibility for studies where a multitude of experimental parameters must be compensated. Based on our previous work, both instrumental designs can provide spectral information of sufficient quality for both measurement objective (noninvasive clinical measurements and feedback control of crystal growth rates). Furthermore, both designs can be miniaturized and both are sufficiently rugged for possible use in space.

Research toward the development of noninvasive measurement technology will focus on the following critical health indicators: hydration, glucose and lactate levels in human subjects. Preliminary results from our laboratory illustrate the potential for measuring each of these analytes noninvasively with NIR spectroscopy. The long-term objective is to provide reagent-less, compact, and low-power instrumentation that is capable of continuous or periodic monitoring of the health status of human subjects during long-term space flights. The intent is to provide a monitor of wellness for periods of sleep and during routine activity, and to provide critical and immediate blood chemistry information during episodes of trauma.

Pertinent to our second objective, a means to monitor and control the rate of crystal growth is important to increasing productivity of microgravity experiments. In addition, the proposed technology will permit control of key experimental parameters (i.e., if we grow protein crystals at the same rate on earth and in space, is the quality of the resulting crystals significantly improved due to microgravity?). Currently, there is no means of measuring and controlling nucleation and growth rates, even approximately, in any of the microgravity experiments. Our efforts will attempt to provide control over the most meaningful variable in protein crystallization, the rate of crystal growth. The need for continuous chemical monitoring is paramount for reaping the technological benefits of the microgravity environment of space and in the long-term exploration of space. For example, the primary objective of this proposal is to develop noninvasive NIR monitors for measuring hydration, glycemia, and blood lactate concentrations. Together, these targeted analytes provide an indication of the overall health status of the human body, particularly in terms of tissue perfusion and cellular function. During long-term space flights, changes in basal levels of hydration and glycemia might indicate a dangerous degradation in the life-support system for the monitored individual. Furthermore, in-flight trauma could result in dehydration, blood loss and burns, which can cause hypovolemic shock. Under these conditions, peripheral water levels drop, blood glucose levels decrease and blood lactate levels increase dramatically (i.e., 10-30 mM) and immediate intravenous perfusion is required to prevent death. The proposed research is the first step in the development of noninvasive continuous sensors for health monitoring for extended periods of space flight.

In addition, the proteome initiative in biomedical science promises to produce tens of thousands of new proteins for which structure and function will need to be determined. X-ray crystallography is the best method for establishing the tertiary structure of these proteins and this technique requires the availability of large, high quality crystals. Experimental evidence suggests that larger and higher quality crystals can be attained in the microgravity environment of space. A major objective of this research effort is to develop noninvasive NIR spectroscopic sensors for monitoring the concentration of dissolved protein as a means to control the rate of crystallization in a temperature-controlled feedback manner. Temperature control offers the potential to grow crystals at the same rate of growth on earth and in space. More importantly, feed-back control of crystallization should enhance the formation of diffraction quality crystals of difficult, membrane-bound proteins.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Research efforts continue to focus on the development of noninvasive sensing technology for monitoring protein crystallization and for detecting clinically relevant molecules in the human body. Progress has centered on the continued development of novel instrumentation for noninvasive measurements, the continued development of an animal model for in vivo clinical measurements, and the exploration of the net analyte signal as a means for robust analytical measurements.
Novel Instrumentation. Instrumentation is being developed for noninvasive measurements of protein crystallization and complex biomedical systems. Three basic optical technologies are under consideration, including acousto-optical tunable filters (AOTF), multiple bandpass filter ssystems (Kromoscopy), and tunable diode lasers.

The AOFT system was described in an earlier report and offers a fully solid-state and rugged spectrometer design. Progress over the last year includes the use of this AOTF system for measuring urea levels in the dialysate stream within a hemodialysis unit. This success demonstrates the analytical utility of this approach. Furthermore, this AOTF unit was deployed for several months in the hemodialysis ward of a hospital. During this period, the instrumentation provided excellent performance, underscoring its durability.

A multiple filter system is under development as a means to further improve the robustness of these systems. This system uses a set of three or four filters to selectively extract analyte information without the need for a fancy spectrometer or complex multivariate calibration model. The key for success is to find filters that emphasize the selective component of the analyte spectrum relative to the sample matrix. In complex samples, finding a suitable set of filters is difficult, but not impossible. In simpler matrixes, such as dialysate samples and protein crystallization media, an effective set of filters is relatively straightforward. Issues of selectivity, limit of detection, sensitivity, and robustness of calibration are under investigation.

The combination region of the near infrared spectrum (4000 – 5000 cm-1) is the best spectral range for noninvasive sensing of organic molecules in aqueous samples. Unfortunately, few light sources are available for this region of the spectrum. Generally, a tungsten filament lamp is used to supply incident radiation for spectroscopic measurements. Such sources generate considerable amounts of heat that must be dissipated and are unstable over time. In addition, tungsten filaments are inefficient over the combination spectral region and typically must be large to generate sufficient radiant powers for high signal-to-noise ratio measurements. We are pursuing the development of a tunable diode laser to overcome the limitations of tungsten sources. Selected semiconductor lasers offer high radiant efficiencies with emission centered at 2.30 μm or 4350 cm-1. The trick is to provide a mechanism for tuning the laser to cover frequencies from

Task Progress:	4500 to 4200 cm-1. We have succeeded in establishing such a tunable laser system that uses an AOTF external cavity. The combination of an AOTF external cavity and the semiconductor laser results in a widely tunable source with copious amounts of radiant powers over the combination wavelengths. Progress includes the use of refractive index matched interfaces to minimize refection losses. This system is currently limited by variations induced by feedback coupling into the laser from the AOTF external cavity. Efforts are ongoing to reduce this mismatch, which will greatly improve stability. In Vivo Measurements. Efforts to measure clinical components noninvasively in human subjects are complex and demands high signal-to-noise ratio spectroscopy coupled with a detailed understanding of the chemistry, physics, and physiology associated with the measurement. For many years our focus has been on in vitro experiments where experimental variables can be controlled and, ultimately, understood. These in vitro experiments have been very informative and have identified many critical experimental parameters that must be controlled or considered before noninvasive in vivo measurements will be successful. As described in our previous report, we are now progressing to in vivo experiments, by establishing an animal model for noninvasive clinical measurements in human subjects. This model consists of harmless transmission spectra collected across a specific region of rat skin. Spectroscopically, this rat skin is very similar to a fold of skin on the back of the hand of tested human subjects. To a first approximation, these spectra are composed of spectral features associated with water, protein, fat, and scattering processes. This animal model is being used to explore the utility of noninvasive clinical measurements for human evaluations. Experiments are ongoing for the noninvasive measurement of glucose, lactate, and urea. We continue to broaden our understanding of the natural sources of spectral variance from the living tissue matrix. Methods are being developed to compensate for these variations, thereby resulting in superior analytical measurements. Lastly, progress continues toward improving the instrumental signal-to-noise for these in vivo experiments. Net Analyte Signal. Over the past year, we have realized the inherent attractiveness of pursuing the net analyte signal (NAS) for noninvasive analytical measurements. The NAS corresponds to that component of the analyte spectrum which is orthogonal to all non-analyte sources of spectral variations. The NAS can be viewed from straightforward linear algebra calculations. Successful analytical measurements demand that the magnitude of the NAS is greater than the underlying noise or uncontrollable spectral variance. Research continues to define the NAS for targeted biomolecules, such as glucose, lactate, and urea. Recent findings demonstrate the NAS for each of these solutes from in vivo spectra collected from the animal model described above. This NAS is solid evidence of the chemical origin of the spectroscopic information used to estimate analyte concentrations. Such NAS signals will serves as a critical guide for future in vivo experiments and instrumentation development.
Bibliography Type:	Description: (Last Updated: 03/14/2008)
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